Governing Equations for Model 502

I. THE BASIC EQUATIONS FOR 502

We inherit the majority of the our equation from model 501. Note in model 502, we define the pressure as the following:

$$P \equiv (\rho - \rho_{imp})(T_e + T_i) + \frac{m_{bg}}{m_{imp}}\rho_{imp}T_i + \langle Z \rangle_{imp}\frac{m_{bg}}{m_{imp}}\rho_{imp}T_e. \tag{1}$$

Here T_e and T_i are the electron temperature and ion temperature respectively. Hence we can define $\alpha_e \equiv \langle Z \rangle_{imp}\frac{m_{bg}}{m_{imp}} - 1$ and $\alpha_i \equiv \frac{m_{bg}}{m_{imp}} - 1$, so that we have

$$P = (\rho + \alpha_i\rho_{imp})T_i + (\rho + \alpha_e\rho_{imp})T_e. \tag{2}$$

Note that in the code, when we do variation of T_e, we have

$$\delta(P)_{T_e} = \rho\delta T_e + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\rho_{imp}\delta T_e. \tag{3}$$

While, for example, the x gradient of P would have the following form

$$\begin{aligned}\partial_x P = {} & (\rho + \alpha_i\rho_{imp})\partial_x T_i + (\partial_x\rho + \alpha_i\partial_x\rho_{imp})T_i \\ & + (\partial_x\rho + \alpha_e\partial_x\rho_{imp})T_e + \left[\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\rho_{imp}\right]\partial_x T_e.\end{aligned} \tag{4}$$

The variation of $\partial_x P$ will then go further

$$\begin{aligned}\delta(\partial_x P)_{T_e} = {} & \left[\partial_x\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\partial_x\rho_{imp}\right]\delta T_e \\ & + \left[\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\rho_{imp}\right]\partial_x\delta T_e \\ & + \left[\left(2\frac{\partial\alpha_e}{\partial T_e} + \frac{\partial^2\alpha_e}{\partial T_e^2}T_e\right)\rho_{imp}\right]\partial_x T_e\delta T_e.\end{aligned} \tag{5}$$

Last, the second gradient of P would be

$$\begin{aligned}\partial_x^2 P_i = {} & (\rho + \alpha_i\rho_{imp})\partial_x^2 T_i + 2(\partial_x\rho + \alpha_i\partial_x\rho_{imp})\partial_x T_i \\ & + \left(\partial_x^2\rho + \alpha_i\partial_x^2\rho_{imp}\right)T_i.\end{aligned} \tag{6}$$

$$\begin{aligned}\partial_y^2 P_i = {} & (\rho + \alpha_i\rho_{imp})\partial_y^2 T_i + 2(\partial_y\rho + \alpha_i\partial_y\rho_{imp})\partial_y T_i \\ & + \left(\partial_y^2\rho + \alpha_i\partial_y^2\rho_{imp}\right)T_i.\end{aligned} \tag{7}$$

$$\partial_x\partial_y P_i = (\rho + \alpha_i\rho_{imp})\,\partial_x\partial_y T_i + (\partial_x\rho + \alpha_i\partial_x\rho_{imp})\,\partial_y T_i + (\partial_y\rho + \alpha_i\partial_y\rho_{imp})\,\partial_x T_i + (\partial_x\partial_y\rho + \alpha_i\partial_x\partial_y\rho_{imp})\,T_i. \tag{8}$$

$$\partial_x^2 P_e = (\rho + \alpha_e\rho_{imp})\,\partial_x^2 T_e + 2\left[\partial_x\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\partial_x\rho_{imp}\right]\partial_x T_e + \left(\partial_x^2\rho + \alpha_i\partial_x^2\rho_{imp}\right)T_e + \left[\left(2\frac{\partial\alpha_e}{\partial T_e} + \frac{\partial^2\alpha_e}{\partial T_e^2}T_e\right)\rho_{imp}\right]\partial_x T_e^2. \tag{9}$$

$$\partial_y^2 P_e = (\rho + \alpha_e\rho_{imp})\,\partial_y^2 T_e + 2\left[\partial_y\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\partial_y\rho_{imp}\right]\partial_y T_e + \left(\partial_y^2\rho + \alpha_i\partial_y^2\rho_{imp}\right)T_e + \left[\left(2\frac{\partial\alpha_e}{\partial T_e} + \frac{\partial^2\alpha_e}{\partial T_e^2}T_e\right)\rho_{imp}\right]\partial_y T_e^2. \tag{10}$$

$$\partial_x\partial_y P_e = (\rho + \alpha_e\rho_{imp})\,\partial_x\partial_y T_e + \left[\partial_x\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\partial_x\rho_{imp}\right]\partial_y T_e + \left[\partial_y\rho + \left(\frac{\partial\alpha_e}{\partial T_e}T_e + \alpha_e\right)\partial_y\rho_{imp}\right]\partial_x T_e + (\partial_x\partial_y\rho + \alpha_i\partial_x\partial_y\rho_{imp})\,T_e + \left[\left(2\frac{\partial\alpha_e}{\partial T_e} + \frac{\partial^2\alpha_e}{\partial T_e^2}T_e\right)\rho_{imp}\right]\partial_x T_e\partial_y T_e. \tag{11}$$

A. The Ampere's Law and the Induction Equation

The induction equation and the Ampere's Law is unchanged

$$\frac{\partial\psi}{\partial t} = \eta\left(T_e\right)\Delta^*\psi - R\left\{u,\psi\right\} - F_0\frac{\partial u}{\partial\phi}, \tag{12}$$

$$j = \Delta^*\psi, \quad j_\phi = -j/R. \tag{13}$$

B. The Vorticity equation

The momentum equation have the following form

$$R\nabla\cdot\left[R^2\left(\rho\nabla_{pol}\frac{\partial u}{\partial t} + \nabla_{pol}u\frac{\partial\rho}{\partial t}\right)\right] = \frac{1}{2}\left\{R^2\left|\nabla_{pol}u\right|^2, R^2\rho\right\} + \left\{R^4\rho\omega, u\right\} - R\nabla\cdot\left[R^2\nabla_{pol}u\nabla\cdot(\rho\mathbf{v})\right] + \{\psi, j\} - \frac{F_0}{R}\frac{\partial j}{\partial\phi} + \left\{P, R^2\right\} + R\mu\left(T_e\right)\nabla_{pol}^2\omega. \tag{14}$$

Here, we again have

$$\omega = \frac{1}{R}\frac{\partial}{\partial R}\left(R\frac{\partial u}{\partial R}\right) + \frac{\partial^2 u}{\partial Z^2}. \tag{15}$$

C. The Total Continuity Equation

Then we move on to the density equation

$$\frac{\partial}{\partial t}\rho = -\nabla\cdot(\rho\mathbf{v}) + \nabla\cdot\left[D_D\nabla\left(\rho-\rho_{imp}\right)\right] + \nabla\cdot\left(D_{imp}\nabla\rho_{imp}\right) + S_D + S_{imp}. \tag{16}$$

Here, D_D and D_{imp} are the diffusion for deuterium and the impurity species respectively, and ρ_{imp} is the total density of impurity species.

D. The ion pressure equation

The major change begins here. We separate the electron and ion pressure equation. For historical reason, the electron pressure equation shall be equation 9, and ion one remains equation 6. Due to the large mass difference between electron and ion, both ohmic heating and radiation (mostly consist of line radiation, recombination radiation and bremsstrahlung radiation) are acting on the electron pressure, not ion.

The ion pressure equation would be

$$\begin{aligned}\frac{\partial}{\partial t}\left[\left(n_{bg}+n_{imp}\right)T_i\right] = &-\mathbf{v}\cdot\nabla\left[\left(n_{bg}+n_{imp}\right)T_i\right] - \gamma\left(n_{bg}+n_{imp}\right)T_i\nabla\cdot\mathbf{v}\\ &+\nabla\cdot\left(\kappa_\perp\nabla_\perp T_i + \kappa_\parallel\nabla_\parallel T_i\right) + \frac{\gamma-1}{2}\mathbf{v}\cdot\mathbf{v}\left(S_D+S_{imp}\right) + n_i\left(\partial_t T_i\right)_e\end{aligned} \tag{17}$$

The second last term derives from the friction force, similar to the kinetic to thermal energy conversion during inelastic collision. In the code, we have

$$\mathbf{v}\cdot\mathbf{v} = v_\parallel^2 B^2 + R^2\left(\nabla_{pol}u\right)^2. \tag{18}$$

Fruther, $n_i\left(\partial_t T_i\right)_e$ represent the energy transfer between the ion and electron species. Consulting NRL formulary, it is easier to simply write down the electron temperature change then invert to get the ion temperature change, assuming energy conservation during the exchange. The electron temperature change due to energy exchange with ions are

$$n_e\left(\partial_t T_e\right)_i = n_e\left(\nu_{e/imp}+\nu_{e/bg}\right)\left(T_i - T_e\right). \tag{19}$$

Here $\nu_{e/imp}$ and $\nu_{e/bg}$ represent the electron collision rate with the impurity species and the background species respectively. For impurity species, considering the summation of all

charge states, we have

$$\nu_{e/imp} = 1.8 \times 10^{-19} \frac{\left(m_e m_{imp}\right)^{1/2} Z_{eff,imp} n_{imp} \ln \Lambda_{e/imp}}{\left(m_e T_i + m_{imp} T_e\right)^{3/2}} s^{-1}, \tag{20}$$

$$Z_{eff,imp} \equiv \sum_i P_{imp}^{i+} i^2. \tag{21}$$

Here P_{imp}^{i+} is the fractional abundance. All values are in cgs unit except the temperature, which is in eV. For background species, we similarly have

$$\nu_{e/bg} = 1.8 \times 10^{-19} \frac{\left(m_e m_{bg}\right)^{1/2} Z_{bg}^2 n_{bg} \ln \Lambda_{e/bg}}{\left(m_e T_i + m_{bg} T_e\right)^{3/2}} s^{-1}. \tag{22}$$

All values are in cgs unit except the temperature, which is in eV. The corresponding ion temperature change due to this energy exchange is then

$$\left(\partial_t T_i\right)_e = -\frac{n_e}{n_{imp} + n_{bg}} \left(\partial_t T_e\right)_i. \tag{23}$$

For implicit scheme, we also need to state the density and temperature gradient of this energy exchange. We have

$$\partial_{T_i} \left(\partial_t T_e\right)_i = \left(\partial_{T_i} \nu_{e/imp} + \partial_{T_i} \nu_{e/bg}\right) \left(T_i - T_e\right) + \nu_{e/imp} + \nu_{e/bg}, \tag{24}$$

$$\partial_{T_e} \left(\partial_t T_e\right)_i = \left(\partial_{T_e} \nu_{e/imp} + \partial_{T_e} \nu_{e/bg}\right) \left(T_i - T_e\right) - \nu_{e/imp} - \nu_{e/bg}, \tag{25}$$

$$\partial_{\rho_{imp}} \left(\partial_t T_e\right)_i = \left(\partial_{\rho_{imp}} \nu_{e/imp} + \partial_{\rho_{imp}} \nu_{e/bg}\right) \left(T_i - T_e\right), \tag{26}$$

$$\partial_{\rho} \left(\partial_t T_e\right)_i = \left(\partial_{\rho} \nu_{e/imp} + \partial_{\rho} \nu_{e/bg}\right) \left(T_i - T_e\right). \tag{27}$$

Here, we have

$$\partial_{T_i} \nu_{e/imp} = -\frac{3}{2} \frac{m_e}{m_e T_i + m_{imp} T_e} \nu_{e/imp}, \tag{28}$$

$$\partial_{T_e} \nu_{e/imp} = \left[-\frac{3}{2} \frac{m_{imp}}{m_e T_i + m_{imp} T_e} + \frac{\partial_{T_e} Z_{eff,imp}}{Z_{eff,imp}}\right] \nu_{e/imp}, \tag{29}$$

$$\partial_{\rho_{imp}} \nu_{e/imp} = \frac{1}{\rho_{imp}} \nu_{e/imp}, \tag{30}$$

$$\partial_{\rho} \nu_{e/imp} = 0. \tag{31}$$

We also have

$$\partial_{T_i} \nu_{e/bg} = -\frac{3}{2} \frac{m_e}{m_e T_i + m_{bg} T_e} \nu_{e/bg}, \tag{32}$$

$$\partial_{T_e} \nu_{e/bg} = -\frac{3}{2} \frac{m_{bg}}{m_e T_i + m_{bg} T_e} \nu_{e/bg}, \tag{33}$$

$$\partial_{\rho_{imp}} \nu_{e/bg} = -\frac{1}{\rho - \rho_{imp}} \nu_{e/bg}, \tag{34}$$

$$\partial_{\rho} \nu_{e/bg} = \frac{1}{\rho - \rho_{imp}} \nu_{e/bg}. \tag{35}$$

Combining the above equations, the LHS matrix terms for the thermalization should be:

$$\partial_{T_i}\left(n_e\partial_t(T_e)_i\right) = \left(\partial_{T_i}\nu_{e/imp} + \partial_{T_i}\nu_{e/bg}\right)\left(T_i - T_e\right)n_e + \left(\nu_{e/imp} + \nu_{e/bg}\right)n_e, \tag{36}$$

$$\begin{aligned}\partial_{T_e}\left(n_e\partial_t(T_e)_i\right) = &\left(\partial_{T_e}\nu_{e/imp} + \partial_{T_e}\nu_{e/bg}\right)\left(T_i - T_e\right)n_e - \left(\nu_{e/imp} + \nu_{e/bg}\right)n_e \\ &+ \left(\nu_{e/imp} + \nu_{e/bg}\right)\left(T_i - T_e\right)\partial_{T_e}n_e,\end{aligned} \tag{37}$$

$$\begin{aligned}\partial_{\rho_{imp}}\left(n_e\partial_t(T_e)_i\right) = &\left(\partial_{\rho_{imp}}\nu_{e/imp} + \partial_{\rho_{imp}}\nu_{e/bg}\right)\left(T_i - T_e\right)n_e \\ &+ \left(\nu_{e/imp} + \nu_{e/bg}\right)\left(T_i - T_e\right)\partial_{\rho_{imp}}n_e,\end{aligned} \tag{38}$$

$$\begin{aligned}\partial_{\rho}\left(n_e\partial_t(T_e)_i\right) = &\left(\partial_{\rho}\nu_{e/imp} + \partial_{\rho}\nu_{e/bg}\right)\left(T_i - T_e\right)n_e \\ &+ \left(\nu_{e/imp} + \nu_{e/bg}\right)\left(T_i - T_e\right)\partial_{\rho}n_e.\end{aligned} \tag{39}$$

Last, for the ion temperature change due to energy transfer

$$\partial_{T_i}\left((n_{imp} + n_{bg})\partial_t(T_i)_e\right) = -\partial_{T_i}\left(n_e\partial_t(T_e)_i\right), \tag{40}$$

$$\partial_{T_e}\left((n_{imp} + n_{bg})\partial_t(T_i)_e\right) = -\partial_{T_e}\left(n_e\partial_t(T_e)_i\right), \tag{41}$$

$$\partial_{\rho_{imp}}\left((n_{imp} + n_{bg})\partial_t(T_i)_e\right) = -\partial_{\rho_{imp}}\left(n_e\partial_t(T_e)_i\right), \tag{42}$$

$$\partial_{\rho}\left((n_{imp} + n_{bg})\partial_t(T_i)_e\right) = -\partial_{\rho}\left(n_e\partial_t(T_e)_i\right). \tag{43}$$

E. Heating by viscous terms

If we are to add the heating by viscous terms, in principle, we should take the momentum equation, time both side with the conjugation of the velocity then the resulting term which corresponded with the viscous term should be our viscous heating.

For code implementation, the perpendicular viscous term is a bit more complicated since the perpendicular equation is written in the second derivative of the flow function (thus we should time the viscous term with the conjugation of the flow function). On the other hand, the parallel viscosity is more relevant to the physics we are concerned (ion heating by Pavel etc.). So we would begin with the parallel viscous heating.

The kinetic energy depletion by such viscous damping is simply:

$$\begin{aligned}P_\mu &= v_\parallel\mu_\parallel B\nabla^2_{pol}\left(v_\parallel B\right) \\ &= \nabla_{pol}\cdot\left[\mu_\parallel v_\parallel B\nabla_{pol}\left(v_\parallel B\right)\right] - \mu_\parallel\left[\nabla_{pol}\left(v_\parallel B\right)\right]^2.\end{aligned} \tag{44}$$

The first term on the RHS is a viscous kinetic energy transport flux, and the second term on the RHS is the conversion of the kinetic energy into thermal energy. Thus the corresponding

heat source in the pressure equation should then simply be $\delta\partial_t P = (\gamma-1)\,\mu_\parallel \left[\nabla_{pol}\left(v_\parallel B\right)\right]^2$. This heating should act on the ion temperature.

F. The electron pressure equation

Similarly, we have the electron pressure equation as

$$\begin{aligned}\frac{\partial}{\partial t}\left(n_e T_e\right) = &-\mathbf{v}\cdot\nabla\left(n_e T_e\right) - \gamma n_e T_e \nabla\cdot\mathbf{v} + \frac{\gamma-1}{R^2}\eta\left(T_e\right) j^2 \\ &+\nabla\cdot\left(\kappa_\perp \nabla_\perp T_e + \kappa_\parallel \nabla_\parallel T_e\right) - (\gamma-1)\, n_e n_{imp} P_{rad}\left(T_e\right) + n_e\left(\partial_t T_e\right)_i . \end{aligned} \tag{45}$$

Here, n_e is calculated via the coronal equilibrium assumption, and $P_{rad}\left(T_e\right)$ is the radiation function assuming coronal equilibrium. The frictional heat source is absent here due to the small mass of electron species.

If we include the energy exchange between the free electron kinetic energy and the ionization potential energy, we simply add a source/sink term in the above equation:

$$\begin{aligned}\frac{\partial}{\partial t}\left(n_e T_e\right) = &-\mathbf{v}\cdot\nabla\left(n_e T_e\right) - \gamma n_e T_e \nabla\cdot\mathbf{v} + \frac{\gamma-1}{R^2}\eta\left(T_e\right) j^2 + S_{ion} \\ &+\nabla\cdot\left(\kappa_\perp \nabla_\perp T_e + \kappa_\parallel \nabla_\parallel T_e\right) - (\gamma-1)\, n_e n_{imp} P_{rad}\left(T_e\right) + n_e\left(\partial_t T_e\right)_i . \end{aligned} \tag{46}$$

The source/sink term S_{ion} can be obtained if we treat the ionization potential energy as a conservative energy with an additional energy exchange term:

$$\partial_t E_{ion} + \nabla\cdot\Gamma_{ion} = -S_{ion}. \tag{47}$$

Meanwhile, we have

$$E_{ion}\left(n_e, T_e\right) \equiv \sum_i P_i\left(n_e, T_e\right) E_i, \quad \partial_t E_{ion} = \sum_i \partial_t P_i\left(n_e, T_e\right) E_i, \tag{48}$$

with E_i being the accumulated ionization energy of charge state i. Essentially the inclusion of the ionization potential energy contribution add three amat terms to the LHS of the equation and two additional terms to the RHS. **Note we have ignored all $\partial E_{ion}/\partial n_e$ terms as those are small for the parameter regime we are considering.**

G. Parallel Velocity Equation

As for the parallel velocity equation, we have the same with model501

$$\begin{aligned}B^2\frac{\partial}{\partial t}\left(\rho v_\parallel\right) = &-\frac{1}{2}\rho\frac{F_0}{R^2}\frac{\partial}{\partial\phi}\left(v_\parallel B\right)^2 - \frac{\rho}{2R}\left\{B^2 v_\parallel^2, \psi\right\} - \frac{F_0}{R^2}\frac{\partial P}{\partial\phi} + \frac{1}{R}\left\{\psi, P\right\} \\ &-B^2\nabla\cdot\left(\rho\mathbf{v}\right) v_\parallel + B^2\mu_\parallel\left(T_e\right)\nabla_{pol}^2 v_\parallel. \end{aligned} \tag{49}$$

H. Impurity Continuity Equation

And ultimately, we have the continuity equation for the impurity species

$$\frac{\partial}{\partial t}\rho_{imp} = -\nabla\cdot(\rho_{imp}\mathbf{v}) + \nabla\cdot(D_{imp}\nabla\rho_{imp}) + S_{imp}. \tag{50}$$

I. Implementation of the Taylor-Galerkin scheme

The general information of the Taylor-Galerkin stabilization in JOREK could be found in Ref. [4]. For reduced MHD, an example of the T-G terms is related to the form of the second and the third term in the RHS in the following continuity equations:

$$\begin{aligned}\frac{\partial\rho}{\partial t} + \mathbf{v}\cdot\nabla\rho = -\rho\nabla\cdot\mathbf{v} + \frac{\delta t}{2}\nabla\cdot\left(v_\parallel^2\mathbf{B}\mathbf{B}\cdot\nabla\rho\right) \\ +\frac{\delta t}{2}\nabla\cdot\left(R^4\nabla u\times\nabla\phi\,(\nabla u\times\nabla\phi)\cdot\nabla\rho\right) + S_\rho.\end{aligned} \tag{51}$$

That is, the convection related with the parallel and the perpendicular velocity. The corresponding weak form would be:

$$\begin{aligned}\int\frac{\partial\rho}{\partial t}\rho^* = -\int\left(\nabla\cdot(\rho\mathbf{v}) + S_\rho\right)\rho^* \\ -\frac{\delta t}{2}\int v_\parallel^2\mathbf{B}\cdot\nabla\rho\mathbf{B}\cdot\nabla\rho^* \\ -\frac{\delta t}{2}\int R^4\left((\nabla u\times\nabla\phi)\cdot\nabla\rho\right)\left((\nabla u\times\nabla\phi)\cdot\nabla\rho^*\right).\end{aligned} \tag{52}$$

For the perpendicular momentum equation, there is only the contribution from the perpendicular velocity, thus the T-G term would have the following form in the non-conservative form:

$$R\nabla\cdot\left[R^2\rho\frac{\partial}{\partial t}\nabla_{pol}u\right] = \cdots + \frac{\delta t}{2}R\nabla\cdot\left[R^2\rho R^2\nabla u\times\nabla\phi\,(\nabla u\times\nabla\phi)\cdot\nabla\omega\right]. \tag{53}$$

Hence the corresponding weak form is

$$TG_{v_E,non-conservative} = -\frac{\delta t}{2}\int\rho R^4\left((\nabla u\times\nabla\phi)\cdot\nabla\omega\right)\left((\nabla u\times\nabla\phi)\cdot\nabla u^*\right). \tag{54}$$

Meanwhile, in the conservative form, it should be

$$TG_{v_E,conservative} = -\frac{\delta t}{2}\int R^4\left((\nabla u\times\nabla\phi)\cdot\nabla\left(\rho\omega\right)\right)\left((\nabla u\times\nabla\phi)\cdot\nabla u^*\right). \tag{55}$$

The additional term in the conservative momentum formula is then corresponding to the gradient of ρ.

For the parallel momentum equation, similarly only the contribution of the parallel velocity is retained. Thus the T-G terms in the non-conservative form is:

$$\rho\frac{\partial}{\partial t}\left(Bv_{\parallel}\right)=\cdots+\rho\frac{\delta t}{2}\nabla\cdot\left(v_{\parallel}^{2}\mathbf{B}\mathbf{B}\cdot\nabla\left(Bv_{\parallel}\right)\right). \tag{56}$$

And the corresponding weak form is:

$$TG_{v_{\parallel},non-conservative}=-\frac{\delta t}{2}\int v_{\parallel}^{2}\mathbf{B}\cdot\nabla\left(Bv_{\parallel}\right)\mathbf{B}\cdot\nabla\left(\rho Bv_{\parallel}^{*}\right). \tag{57}$$

For the conservative form, the weak form changes into

$$TG_{v_{\parallel},conservative}=-\frac{\delta t}{2}\int v_{\parallel}^{2}\mathbf{B}\cdot\nabla\left(\rho Bv_{\parallel}\right)\mathbf{B}\cdot\nabla\left(Bv_{\parallel}^{*}\right). \tag{58}$$

II. THE MEMORY ARRAY

The long-tail effect of the hot electrons arise from the fact that the hot electrons take remarkable time to thermalize with the newly ablated/ionized species compared with the fly time of the fragment, so that the fragment should not see a Maxwellian plasma (which was assumed in all current ablation models), but would see the contribution of this still hot electron species instead. As the ablation rate goes up with $T_e^{5/3}$, this could result in significant underestimation of ablation for current model.

On the other hand, the preferential depletion of hot electrons by collision with the solid pellet fragments themselves would result in the so-called "self-limiting effect" which tend to reduce the aforementioned long tail[3]. For a ITER plasma, it can be estimated that when injecting on the order of $\mathcal{O}\left(10^{24}\right)$ deuterium (along with mixed impurity), the self-limiting time is somewhat smaller but comparable with the fragment fly time across a flux surface with width $1cm$. Thus this hot tail effect can be expected to be somewhat mitigated but still important.

To represent such a process, we consider the so-called memory array approach. We save the temperature and density of the hot tail electrons as a function of the magnetic flux ψ. At each time, for each fragment, we search for ψ, then use ψ of each fragment to find the temperature and density of the hot tail. Then, we choose the total ablation rate to be

$$G_{total}=\max\left(G_{bg},G_{ht}\right). \tag{59}$$

For each time step, the hot electron density is updated according to the self-limiting effect:

$$\partial_t N_{e,ht} = -2 \sum_i n_{e,ht} v_{th} \pi r_p^2. \tag{60}$$

Here $N_{e,ht}$ is the total amount of hot electrons within the "section" of flux surfaces, while $n_{e,ht}$ is the hot electron density, v_{th} is the hot electron thermal velocity.

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